

Power Transformer Differential Protection Based On Optimal Probabilistic Neural Network

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Abstract—In this paper, the optimal probabilistic neural network (PNN) is proposed as the core classifier to discriminate between the magnetizing inrush and the internal fault of a power transformer. The particle swarm optimization is used to obtain an optimal smoothing factor of PNN which is a crucial parameter for PNN. An algorithm has been developed around the theme of the conventional differential protection of the transformer. It makes use of the ratio of voltage-to-frequency and amplitude of differential current for the determination of operating condition of the transformer. The performance of the proposed heteroscedastic-type PNN is investigated with the conventional homoscedastic-type PNN, feedforward back propagation (FFBP) neural network, and the conventional harmonic restraint method. To evaluate the developed algorithm, relaying signals for various operating condition of the transformer, including internal and external faults, are obtained by modeling the transformer in PSCAD/EMTDC. The protection algorithm is implemented by using MATLAB.

Index Terms—Artificial neural network (ANN), digital differential power transformer protection, particle swarm optimization, probabilistic neural network, protective relaying.

I. INTRODUCTION

MEDIUM and large power transformers are very important and vital component of electric power systems. Due to its importance and cost, its protection needs to be addressed properly. The protection should be fast and reliable. Proper continuous monitoring of power transformer can provide early warning of electrical failure and can prevent catastrophic losses. It can minimize damages and enhanced the reliability of power supply. Accordingly, high expectations are imposed on power transformer protective relays. Expectations from protective relays include dependability (no missing operations), security (no false tripping), speed of operation (short fault clearing time) and stability. Differential relaying principle is used for protection of medium and large power transformers. This superior approach compares the currents at all terminals of the protected transformer by computing and monitoring a differential (unbalance) current. The value of differential current greater than no-load value indicates an internal fault. Magnetizing inrush

occurs in transformer at the time of large change in voltages, whenever, polarity and magnitude of residual flux do not agree with polarity and magnitude of ideal instantaneous value of steady-state flux. When there is large and sudden change in the input terminal voltage of transformer, either due to switching-in or due to recovery from external fault, a large current is drawn by the transformer from supply. Similar condition occurs when transformer is energized in parallel with a transformer that is already in service, known as “*sympathetic inrush*” condition. This results in core of transformer getting saturated. This phenomenon is known as magnetizing inrush or in other words, inrush can be described by a condition of large differential current occurring from either the transformer is just switched-in or the system recovers from an external fault or a transformer in energized in parallel to already operated transformer. Magnetizing inrush current may be as high of the order of 10 times of full load current [1]. This resulting high differential current may cause the relay to operate. To avoid the mal-operation of relay, discrimination between magnetizing inrush current and fault current is required. Generally, two approaches are applied to discriminate between magnetizing inrush condition and fault condition [2]. These are harmonic restraint concept and waveform identification concept. The harmonic restraint is based on the fact that the second-harmonic (sometimes the fifth) component of the magnetizing inrush current is considerably larger than in a typical fault current. The literature reveals that the first method, based on harmonic restraint, has been used extensively [2], [3]. The harmonic restraint-based method sometime fails to prevent false tripping of relays because high second-harmonic components are generated during internal faults and low second-harmonic component generated during magnetizing inrush having modern core material of power transformer [4], [5]. Therefore, the detection of second/fifth harmonic cannot be taken as a sufficient index to discriminate between the magnetizing inrush and fault condition of power transformer. The second method consists of distinguishing magnetizing inrush and over-excitation condition from internal fault condition on the basis of waveform identification. This method was carried out by utilizing the differential current peaks, dead angle, and the length of time intervals during which the differential current is near zero [2], [6], [7].

More recently with the developments in artificial neural network (ANN) and Wavelet analysis, their application for protection of power transformer is also getting momentum. In 2000, Xiaoxu Ma *et al.* proposed an algorithm based on the hidden Markov model to discriminate between fault and magnetizing inrush conditions [8]. Wavelet-transform-based methods have better ability of time-frequency location. Their disadvantages are that they need long data window and are

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also sensitive to noise and unpredicted disturbances, which limit their application in relaying [9]. Development of ANN enhances the scope of waveform identification approach. ANN approach is faster, robust and easier to implement than the conventional waveform approach. Since early nineties, ANN is being used in the field of power system protection due to its good generalization ability and learning stability with different topologies. Most of the author used multilayer feedforward neural network (MFFNN) with back propagation learning technique [2], [10]. In the literature, another ANN model called the radial basis function neural network (RBFNN) has been reported for power transformer protection [11].

In the present paper, another type of ANN model, the probabilistic neural network (PNN), is investigated for the protection of power transformer. In differential relaying protection scheme the PNN is utilized as classifier to discriminate between magnetizing inrush and internal fault of power transformer. The performance of PNN is influenced only by spread parameter or smoothing factor i.e., single parameter whereas the performance of MFFNN and RBFNN are influenced by two or more parameters. Hence, an outstanding issue arises to get an optimal smoothing factor for PNN. In this paper, two methods are proposed to achieve the optimal smoothing factor. These methods are particle swarm optimization (PSO) and the conventional method. The selection of PSO is because of its efficiency in solving a plethora of applications in sciences and engineering [12], [13]. In this paper, emphasis is placed on the detection of magnetizing inrush and internal fault recognition of power transformer which is crucial in differential protection scheme. A differential relay algorithm is presented around the theme of the conventional differential protection using PNN classification ability. The classification accuracy of optimal PNNs (heteroscedastic and homoscedastic types), FFBP neural network and the harmonic restraint method based on discrete Fourier transform (DFT) are compared.

II. PROBABILISTIC NEURAL NETWORK (PNN)

PNN is a kind of feedforward neural network. The original PNN structure is a direct neural network implementation of Parzen nonparametric probability density function (pdf) estimation and Bayes classification rule [14], [15]. The standard training procedure of PNN requires a single pass-over all the patterns of the training set [14]. This characteristic renders PNN faster to train as compared to feedforward back propagation (FFBP) neural network [16]. The only draw back of PNN is requirement of larger storage for exemplar patterns. As computer memory has become very cheap and effective, the cost and size of larger storage are no longer concern these days. PNN is widely used in the area of pattern recognition, nonlinear mapping, fault detection and classification, estimation of probability of class membership and likelihood ratios [17].

PNN can be constructed in a number of ways. Specht [14] used all training patterns as the centers of Gaussian kernel functions and assumed a common variance or co-variance (called homoscedastic PNN). Because variance of homoscedastic PNN cannot be determined analytically, it requires a training phase, followed by a validation phase, before it can be used in a testing phase. To avoid the use of a validation data set, another type

known as heteroscedastic PNN (uncorrelated and separate variance) is used. However, there is an outstanding issue associated with PNNs that is determination of optimal smoothing parameter which plays a crucial role in PNN classifier, and is often data dependent. The heteroscedastic type of PNN is more difficult to train, using maximum likelihood (ML) training algorithm, because of numerical difficulties [18]. In the present paper, a simple and robust optimization technique has been used to solve this type of numerical problem by using PSO which is heuristic optimization technique [12].

The PNN structure is shown in Fig. 1. It is a four-layer feedforward neural network that is capable of realizing or approximating the optimal classifier. In PNN generally Gaussian activation function is used because it behaves well and is easily computed. It is not based on normal distribution. The activation function for PNN is derived from estimates of pdf based on the training patterns as following [19]:

Let $X \in R^d$ be a d -dimensional pattern vectors and its associated class be $i \in (S_1, S_2, S_3, \dots, S_k)$, where k is the number of possible classes. If a *posteriori* probability $P_r(S_i/x)$ that is from class S_i is by Bayes' rule

$$P_r(S_i/x) = \frac{P_r(x/S_i)P_r(S_i)}{P(x)} \quad (1)$$

where $P_r(x/S_i)$, $i = 1, 2, 3, \dots, k$ is a *a priori* pdf of the pattern in classes to be separated.

$P_r(S_i)$, $i = 1, 2, 3, \dots, k$ are *a priori* probabilities of the classes. $P(x)$ is assumed to be constant.

The decision rule is to select class S_i for which $P_r(S_i/x)$ is maximum. This will happen if for all $j \neq i$

$$P(x/S_i)P_r(S_i) > P(x/S_j)P_r(S_j). \quad (2)$$

It is assumed that the *a priori* probabilities $P_r(S_i)$ of the classes are known and *a priori* pdf, $P(x/S_i)$ is Gaussian then the estimator for the *a priori* pdf is

$$\hat{P}(x/S_i) = \frac{1}{(2\pi)^{d/2}\sigma_i^d|S_i|} \sum_{j=1}^{n_i} \exp \times \left[\frac{-(x - x_j)^T(x - x_j)}{2\sigma_i^2} \right] \quad (3)$$

where

x_j^i	j th exemplar pattern from class S_i ;
$ S_i = n_i $	cardinality of the set patterns in class S_i ;
σ_i	smoothing factor.

The input layer has d units, to which the d -dimensional input vector $X \in R^d$ is applied. The first hidden layer has one pattern unit for each pattern exemplar. Therefore, each such pattern unit may be associated with a generic term depicted in the summation of (3) for the i th class. The second hidden layer contains one summation unit for each class. The output layer is decision layer used for implementing the decision rule by selecting maximum posteriori probability, $P_r(S_i/x)$ from outputs preceding summation layer for each i . The network is constructed by setting weight vector to one of the pattern unit equal to each distinct pattern vector in the training set from a certain class and then connecting the outputs of the pattern units to the appropriate

summation units for that class. In present paper, homoscedastic and heteroscedastic-type PNNs are realized by using PSO. Homoscedastic-type PNN is constructed by the conventional trial and error method, as calculation of smoothing factor becomes very simple, thereby avoiding complex calculations [19]. The procedure is given as

$$\sigma_i = g d_{j_{avg}}^i \quad (4)$$

where

- d_j^i distance between the j th exemplar pattern and nearest exemplar pattern in class i ;
- g constant that has been found experimentally.

III. PARTICLE SWARM OPTIMIZATION (PSO)

PSO is a population based stochastic optimization technique developed by Eberhart and Kennedy in 1995 [12], inspired by social behavior of birds flocking or fish schooling. The main advantages of PSO algorithm are simple concept, easy implementation, robustness to control parameters and computational efficiency when compared with mathematical algorithm and other heuristic optimization techniques. PSO is designed and proved to be very effective in solving real valued global optimization problems [13]. Moreover, it does not require gradient information of the objective function under consideration, but only its values, and it uses only primitive mathematical operators.

In PSO, population is called swarm and individuals (i.e., the points) are called particles. Each particle moves with an adaptable velocity within search space and retains in a memory the best position it ever encountered. This best position is shared with other particles in the swarm at each iteration. Two variants of the PSO algorithm were developed [13], one with a global neighborhood, and other with a local neighborhood. According to the global variant, each particle moves towards its best previous position and towards the best particle in the whole swarm, whereas according to the local variant, each particle moves towards its best previous position and towards the best particle in its restricted neighborhood [13]. In general, the global variant of PSO exhibits faster convergence rates, although, in some cases it may reduce the swarm's diversity very fast, thereby getting trapped in local minimizers. On other hand, the local variant, especially when the neighborhood size is small, exhibits superior exploration capabilities at the cost of slower convergence.

Assuming an n -dimensional search space, the position and velocity of individual i are represented as the vector $Z_i = (z_{i1}, z_{i2}, z_{i3}, \dots, z_{in})$ and $V_i = (v_{i1}, v_{i2}, v_{i3}, \dots, v_{in})$, respectively, in the PSO algorithm.

Let the best previous position $BP_i = (bp_{i1}, bp_{i2}, bp_{i3}, \dots, bp_{in})$ and $GB_i = (gb_{i1}, gb_{i2}, gb_{i3}, \dots, gb_{in})$, respectively be the best position of individual i and its neighbor's best position so far. The updated velocity of individual i is modified by using following equations:

$$V_i^{k+1} = w V_i^k + c1 r1 (BP_i^k - Z_i^k)$$

$$+ c2 r2 (GB_i^k - Z_i^k) \quad (5)$$

$$Z_i^{k+1} = Z_i^k + V_i^{k+1} \quad (6)$$

where

- V_i^k velocity of individual i at instant k ;
- w weight parameter;
- $c1, c2$ two positive constants called cognitive and social;
- $r1, r2$ random number between 0 and 1;
- Z_i^k position of individual i until iteration;
- BP_i^k best position of individual i until iteration;
- GB_i^k best position of the group until iteration.

Performance of each particle is measured according to a pre-defined fitness function which is problem dependant. The inertia weight w is employed to control the impact of previous history of velocities on the current velocity. A large inertia weight w facilitates global exploration (searching new areas) while a smaller inertia weight tends to facilitate local exploration to fine-tune the current search area. The inertia weight w can be set to the following equation [13]:

$$W = W_{\max} - \frac{W_{\max} - W_{\min}}{\text{iter}_{\max}} \times \text{iter} \quad (7)$$

where

- $w_{\min}$ 0.9, $w_{\max} = 0.4$;
- $\text{iter}_{\max}$ maximum iteration number;
- iter current iteration number.

A pseudocode of PSO to obtain optimal smoothing factor for heteroscedastic type of PNN is given as follows:

FOR each Particle

Initialize Particle

End

Do

FOR each Particle

Compute fitness function $f((\sigma_i^{k+1}))$ by the leave-one-out misclassification proportion on training exemplar pattern set.

If fitness value is better than the best fitness (BP) in history

Set current value as the new BP

END

Choose the Particle with the best fitness as the GB

FOR each Particle

Compute its velocity

Update its position (i.e., smoothing factor $f((\sigma_i))$)

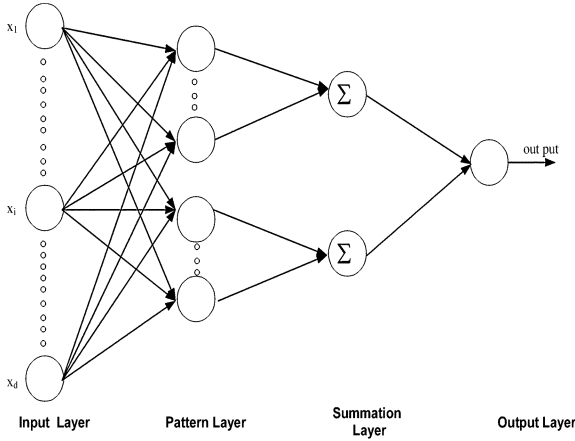


Fig. 1. PNN structure.

END

WHILE maximum iterations or stop criteria are not attained.

Leave-one-out (LOO) error estimation is an important statistical tool for assessing generalization performance [20]. In the PSO method, a swarm of particle (i.e., smoothing parameter σ) is initialized randomly in $[0, 1]^n$, n denotes the dimension of optimization problem. The LOO misclassification proportion on the training set is computed and this value is used as the fitness value i.e., $f((\sigma_i))$. In LOO method, a PNN is trained using all but one of patterns from the training set. The executed pattern is subsequently used to assess the classification ability of the network. This process is repeated excluding a different pattern of the training set each time, until all patterns of this set are executed once. This adaptation process is terminated when maximum number of iteration is reached.

IV. SIMULATION AND TRAINING CASES

During power transformer operation it encounters any one of the following conditions.

- Normal condition.
- Magnetizing inrush/Sympathetic inrush condition.
- Over-excitation condition.
- Internal fault condition.
- External fault condition.

In normal condition, rated or less current flows through the transformer. In this condition normalized differential current is almost zero (only no load component of current). Whenever, there is large and sudden change in the input terminal voltage of transformer, either due to switching-in or due to recovery from external fault, a large current is drawn by the transformer from the supply. As a result, the core of transformer gets saturated. This phenomenon is known as magnetizing inrush. Magnetizing inrush can occur in an already energized transformer when a nearby transformer is energized. A common situation of sympathetic inrush is encountered when a transformer is energized

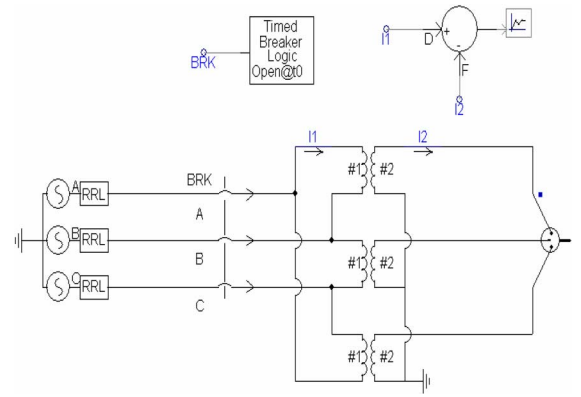


Fig. 2. Simulation diagram of magnetizing inrush under no-load condition of transformer.

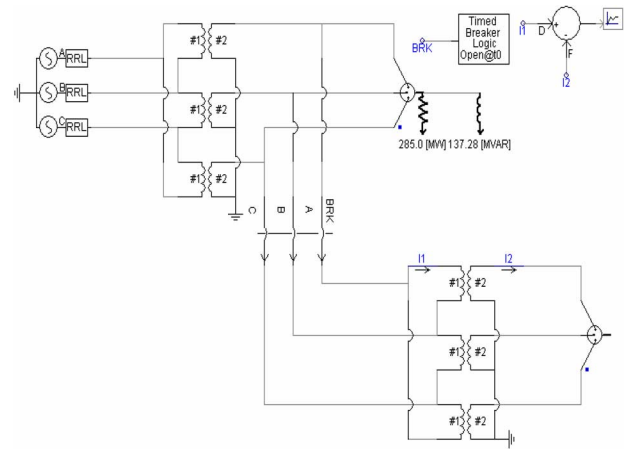


Fig. 3. Simulation diagram of sympathetic inrush condition of transformer.

in parallel with another transformer already in service. The phenomenon which causes inrush current to flow in a previously energized transformer is known as the “sympathetic inrush”. As the paralleled transformer is being energized by closing the breaker, an inrush current is established in the primary of this transformer and this inrush current has dc component. The dc component of the inrush current can also saturate the already energized transformer, resulting in an apparent inrush current. This transient current, when added to the current of already energized transformer, results in an asymmetrical current that is very low in harmonics. This would be the current flowing in the supply circuit to both transformers. Sympathetic inrush current may not have sufficient amount of the second harmonic in it to prevent the relay from tripping. Sympathetic inrush current depends on same factors on which switching-in and recovery from fault magnetizing inrush current depends.

PSCAD/EMTDC software is used to generate training and test signals under different operating conditions of transformer that are normal, over-excitation, magnetizing inrush, sympathetic inrush, and fault (phase-to-phase, phase-to-ground, and external fault). The simulation set up is given in Figs. 2–5.

While simulating magnetizing inrush condition, switching-in angle, remanent flux in the core and load condition are considered as the magnitude and the wave-shape of magnetizing inrush current depends on these factors. Energization angle

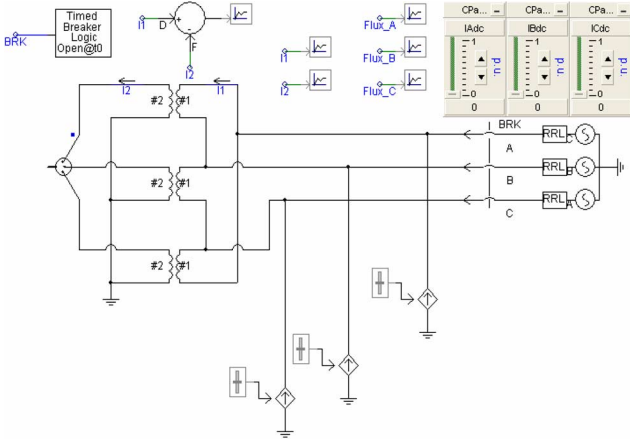


Fig. 4. Simulation diagram of magnetizing inrush under no-load considering remanence flux in transformer.

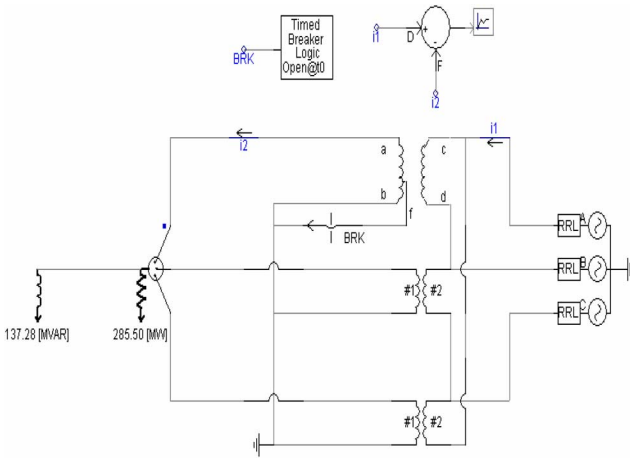


Fig. 5. Simulation diagram of phase-to-ground fault under full-load condition of transformer.

is varied from 0 to 360 degrees in interval of 30 degrees and remanent flux is considered 0% to 80% of the peak flux linkages generated at rated voltage with no load and full load conditions to generate training signals whereas, in case of testing signals energization angle is varied in interval of 15 degrees. The desired remanence can be set in unenergized transformer with controlled dc current sources in PSCAD/EMTDC simulation model [21]. For internal fault, training and testing is done by simulating fault by varying number of winding turns short circuited from 1% to 99% of power transformer winding turns. Phase-to-ground fault at different locations such as 5%, 15%, 25%, 40%, and 50% from the neutral end of the winding as well as terminal fault are simulated. The detailed information of power transformer PSCAD/EMTDC simulation model to simulate internal fault (Fig. 14) is given in the Appendix I. Three-phase transformer of 315 MVA at 400/220 kV, 200 MVA at 220/110 kV, and 160 MVA at 132/220 kV are modeled by using PSCAD/EMTDC. For the simulation of these transformers through PSCAD/EMTDC, the realistic data obtained from M. P. State Electricity Board, Jabalpur, India is used. The test signals so acquired by simulating various operating conditions of transformer are shown in Figs. 6–10. The simulation

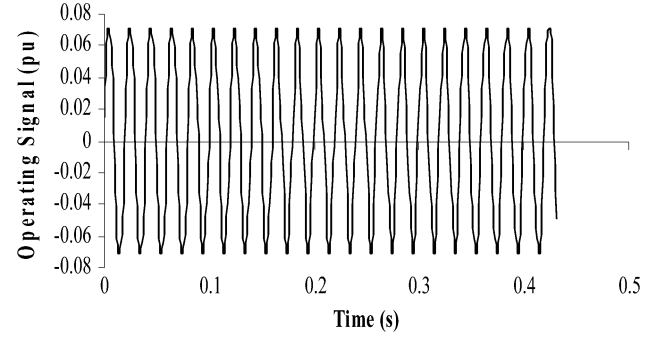


Fig. 6. Typical differential current waveform under normal operation.

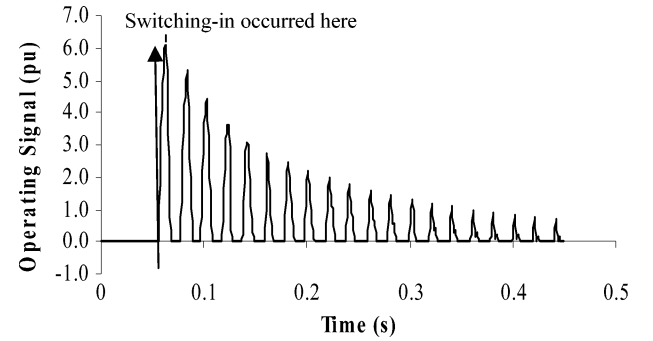


Fig. 7. Typical magnetizing inrush current waveform.

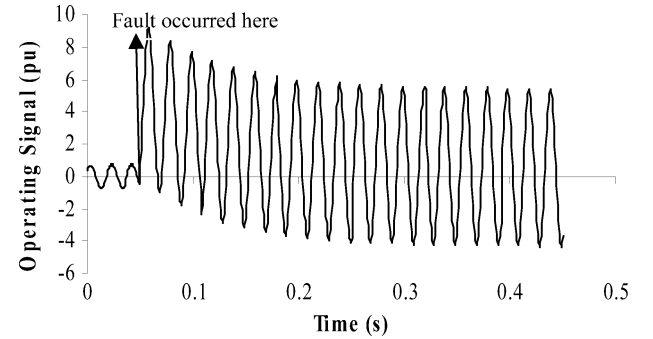


Fig. 8. Typical phase-to-ground fault current waveform.

was done at the rate of 12 samples per cycle of 50 Hz ac supply in view of reported experience on different digital relay designs [22]. The developed protection algorithm was implemented in MATLAB.

V. IMPLEMENTATION OF PNN BASE ALGORITHM AND RESULTS

In proposed PNN architecture, four layered structure is used as shown in Fig. 1. In input layer 12 neurons, in the first hidden layer 777 neurons, second hidden layer two neurons and at output layer one neuron is considered. The number of neurons in input layer are decided based on dimension of the feature space, i.e., 12 samples per cycle whereas, the number neurons of first hidden layer are decided on the basis of total number of exemplar pattern set used to construct the PNN model. In exemplar pattern set, 444 patterns of magnetizing inrush and 333 patterns of faults are used. Second hidden layer has two

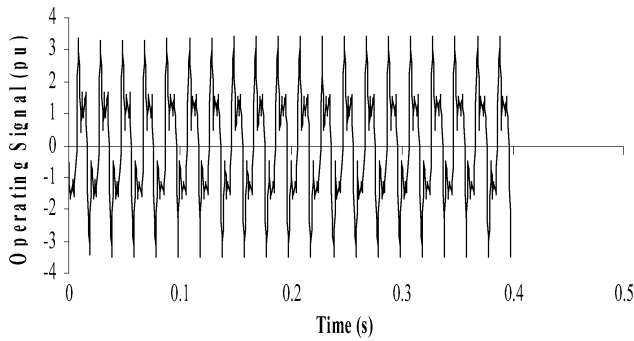


Fig. 9. Typical differential current waveform for over-excitation.

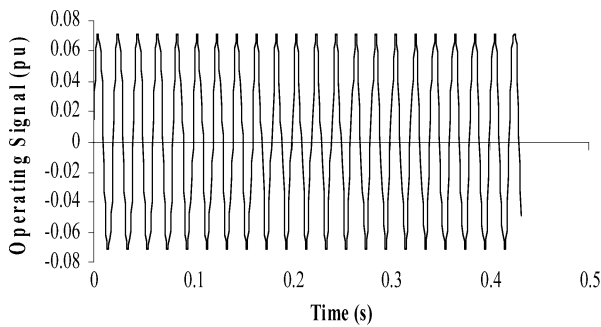


Fig. 10. Typical differential current waveform under external fault condition.

neurons as there are only two numbers of classes to be discriminated i.e., fault or no fault. In output layer one neuron is used, as it is the decision layer used for selecting the maximum *a posteriori* probability, from the outputs of the summation layer. In present work, optimal smoothing factor which is crucial for the classification accuracy of PNN is obtained either by PSO or the conventional method given by (4). By trial and error method the optimal value of multiplying factor (g) is obtained and hence optimal smoothing factor is achieved.

The local variant of PSO algorithm is used, as it exhibit better performance, compared to the global variant, due to its enhanced exploration capability. According to pseudocode given in Section III, PSO algorithm is initialized randomly with swarm of 20 particles. The typical range for the number of particles ranges from 20 to 40. The values of cognitive and social are taken as 2. The maximum number of iteration was set to 100 and inertia weight w is initially set at 0.9. In order to reduce this weight over the iterations, allowing the algorithm to exploit some specific areas, (7) is used. The particles are constrained in the range of 0 to 1, since smoothing factor lies in this range for the normalized data. The LOO strategy is applied for minimizing misclassification error on the training set, while misclassification error on the validation set is monitored after each iteration of the algorithm.

The signal was sampled at the rate of 12 samples per cycle (over a data window of one cycle). Out of 925 sets of data (patterns), 777 patterns sets are used to construct of PNN with optimal smoothing factor which is already obtained by PSO or the conventional method and remaining 148 sets are used to test the network's generalization ability that are different

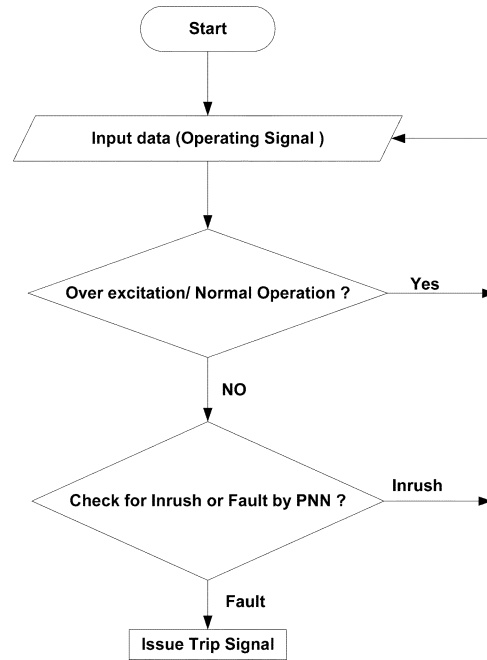


Fig. 11. Flowchart of proposed algorithm.

than patterns used to train the network. These 148 test exemplar pattern sets contain internal fault and magnetizing inrush condition only as these two conditions are very difficult to discriminate as compare to other operating conditions like external fault, over-excitation and normal condition. From a protection point of view, only these two conditions are necessary to identify as the relay has to give trip signal in case of internal fault condition only whereas in other conditions it should not operate. The discrimination between external fault and normal operating condition is made by comparing two consecutive peaks of operating signal. The over-excitation condition is determined by comparing voltage-to-frequency ratio with the rated voltage-to-frequency ratio. If these condition do not exist then magnetizing inrush and internal fault condition is checked by PNN. The PNN gives tripping signal if internal fault condition is found. The flowchart (Fig. 11) clearly indicates these steps for discriminating different operating conditions of power transformer. For different conditions of the test set, fault current magnitude, remanent flux, load condition and switching-in angle were changed to investigate the effects of these factors on the performance of the PNN model. The wave-shape of magnetizing inrush current changed with variation of switching-in instant of transformer which is varied between 0 to 360 degrees. Due to remanence flux, the magnitude of magnetizing inrush current may rise up to two to six times of magnetizing inrush current without remanence effect although the wave-shape remains same. It is found that the PNN classifier based relay is stable even with such high magnitude of magnetizing inrush current caused by remanence flux whereas the conventional harmonic-based relay may mal-operate due to such high magnitude magnetizing inrush current.

Tables IA, IIA, IIIA, and IVA illustrate the number of false detection i.e., number of false positive and false negative while using PNNs (heteroscedastic and homoscedastic types) and

TABLE IA
NUMBER OF FALSE DETECTIONS IN HETEROSCEDASTIC TYPE OF PNN BY PSO

Training transformer ratings	Tested transformer ratings																	
	315 MVA						200 MVA						160 MVA					
	False positives		False negatives				False positives		False negatives				False positives		False negatives			
	Inrush		Faults				Inrush		Faults				Inrush		Phase-to-ground fault			Phase-to-phase fault
	Switching -in angles		Phase-to-ground fault		Phase-to-phase fault		Switching -in angles		Phase-to-ground fault		Phase-to-phase fault		Switching -in angles		Phase-to-ground fault		Phase-to-phase fault	
	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *
315 MVA	0	-	0	-	0	-	0	-	1	2	0	-	0	-	0	-	0	-
200 MVA	0	-	0	-	0	-	0	-	0	-	0	-	0	-	2	2,5	0	-
160 MVA	0	-	1	2	0	-	0	-	2	2,5	0	-	0	-	0	-	0	-

TABLE IIA
NUMBER OF FALSE DETECTIONS IN HOMOSCEDASTIC-TYPE PNN BY TRIAL AND ERROR

Training transformer ratings	Tested transformer ratings																	
	315 MVA						200 MVA						160 MVA					
	False positives		False negatives				False positives		False negatives				False positives		False negatives			
	Inrush		Faults				Inrush		Faults				Inrush		Faults			
	Switching -in angles		Phase-to-ground fault		Phase-to-phase fault		Switching-in angles		Phase-to-ground fault		Phase-to-Phase fault		Switching-in angles		Phase-to-ground fault		Phase-to-phase fault	
	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *
315 MVA	0	-	0	-	0	-	4	15, 125, 305, 310	2	2, 5	1	5	0	-	0	-	0	-
200 MVA	6	15, 45, 125, 130, 305, 315	2	2, 5	2	2, 5	0	-	0	-	0	-	6	15, 45, 125, 130, 305, 310	5	2, 5, 10, 15, 20	4	2, 5, 10, 15
160 MVA	6	15, 45, 125, 130, 305, 310	4	2, 5, 10, 15	2	2, 5	3	125, 305, 310	2	2, 5	1	5	0	-	0	-	0	-

% * represents location of winding from the neutral end at which fault occurs
% ** represents number of recurrences

FFBP neural network as core classifier respectively. Table IA and Table IIIA demonstrates that in the heteroscedastic-type PNN with PSO, there is no false positive detection and very few false negative detections as compare to the homoscedastic-type PNN with PSO. From Tables IIA and Table IIIA, it is clear that in the homoscedastic-type PNN with PSO has less number of false positive and false negative detections as compare to the homoscedastic-type PNN with trial and error method. Whereas in case of a FFBP neural network, more number of false positive and false negative are detected (as given in Table IVA) compared to heteroscedastic-type PNN with PSO

Tables IB, IIB, and IIIB shows the classification accuracy of optimal PNNs (heteroscedastic and homoscedastic-type) classifier by using (8) while Table IVB presents the improvements in classification accuracy by using optimal heteroscedastic and homoscedastic type of PNN. Table IIC illustrates optimal multiplying factor obtained by (4) used to design homoscedastic-type PNN by trial and error method. It successfully discriminate all patterns which is simulated by considering all influencing factors. From the results, it is clear that when power transformer is trained and tested with same rating of transformers than 100% classification accuracy is obtained. But when a transformer is trained and tested on different ratings of transformer than some classification error occurs. To minimize the error and improve the performance of PNN, optimal value of smoothing factor obtained by PSO is considered, it is evident that the classification accuracy improves as illustrated in Table IVB. It is concluded that the conventional trial and error method is more time consuming to set the optimal spread parameter of PNN than the PSO method. In comparison of classification accuracy of heteroscedastic and homoscedastic type of PNN, the overall result of heteroscedastic-type PNN is better than homoscedastic type

$$\text{Classification Error (in\%)} = \frac{\text{No. of False Positive} + \text{No. of False Negative}}{\text{Total Number of Test Cases}} \times 100$$

$$\text{Classification Accuracy (in\%)} = 100 - \text{Classification Error (in\%)}. \quad (8)$$

TABLE IIIA
NUMBER OF FALSE DETECTIONS IN HOMOSCEDASTIC-TYPE PNN BY PSO

Training transformer ratings	Tested transformer ratings																	
	315 MVA						200 MVA						160 MVA					
	False positives		False negatives				False positives		False negatives				False positives		False negatives			
	Inrush		Faults				Inrush		Faults				Inrush		Faults			
	Switching-in angles		Phase-to-ground fault		Phase-to - phase fault		Switching-in angles		Phase-to-ground fault		Phase-to phase fault		Switching-in angles		Phase-to-ground fault		Phase-to-phase fault	
	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *
315 MVA	0	-	0	-	0	-	2	125, 305	1	2	1	5	1	125	0	-	0	-
200 MVA	1	125	1	2	0	-	0	-	0	-	0	-	2	125, 305	2	2, 5	2	5, 10
160 MVA	2	125, 305	1	2	1	5	2	125, 305	1	2	1	5	0	-	0	-	0	-

TABLE IVA
NUMBER OF FALSE DETECTIONS IN FFBP NEURAL NETWORK

Training transformer ratings	Tested transformer ratings																	
	315 MVA						200 MVA						160 MVA					
	False positives		False negatives				False positives		False negatives				False positives		False negatives			
	Inrush		Faults				Inrush		Faults				Inrush		Faults			
	Switching -in angles		Phase-to-ground fault		Phase-to-phase fault		Switching -in angles		Phase-to-ground fault		Phase-to- phase fault		Switching -in angles		Phase-to-ground fault		Phase-to-phase fault	
	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *	% **	Degrees	% **	% *	% **	% *
315 MVA	0	-	1	2	-	-	3	125, 315, 310	3	2, 5, 10	2	5, 10	0	-	2	2, 5	0	-
200 MVA	2	125, 305	2	2, 5	1	5	1	125	0	-	0	-	1	125	1	2	0	-
160 MVA	0	-	1	2	1	5	1	125	3	2, 5, 10	1	5	0	-	-	-	0	-

% * represents location of winding from the neutral end at which fault occurs
% ** represents number of recurrences

TABLE IIB
CLASSIFICATION ACCURACY (IN%) OF HETEROSCEDASTIC-TYPE PNN BY PSO

Trained Transformer Ratings	Tested Transformer Ratings		
	315 MVA	200 MVA	160 MVA
315 MVA	100	99.3	100
200 MVA	100	100	98.6
160 MVA	99.3	98.6	100

TABLE IIB
CLASSIFICATION ACCURACY (IN%) OF HOMOSCEDASTIC-TYPE PNN BY TRIAL AND ERROR

Trained Transformer Ratings	Tested Transformer Ratings		
	315MVA	200 MVA	160 MVA
315 MVA	100	96.0	100
200 MVA	93.2	100	89.8
160 MVA	91.9	95.9	100

of PNN because in the heteroscedastic type of PNN each kernel has its own different variance which minimizes the overlapping

TABLE IIC
OPTIMAL MULTIPLYING FACTOR (G)

Trained Transformer Ratings	Tested Transformer Ratings		
	315 MVA	200 MVA	160 MVA
315 MVA	1.1	1.4	1.1
200 MVA	0.5	0.3	0.5
160 MVA	0.3	0.3	1.1

TABLE IIIB
CLASSIFICATION ACCURACY (IN%) OF HOMOSCEDASTIC-TYPE PNN BY PSO

Trained Transformer Ratings	Tested Transformer Ratings		
	315 MVA	200MVA	160 MVA
315 MVA	100	97.2	99.3
200 MVA	98.6	100	95.9
160 MVA	97.2	97.2	100

of nearest neighbors of different classes to preserve local properties, as well as maximize the generalization ability.

As it is well known, that classification time is an important parameter of relay operation. Several approaches may be found in the literature which claim different discrimination time such

TABLE IVB
IMPROVEMENT IN CLASSIFICATION ACCURACY (IN%) WITH HOMOSCEDASTIC
AND HETEROSCEDASTIC-TYPE PNN BY PSO

Training transformer ratings	Tested transformer ratings								
	315 MVA			200 MVA			160 MVA		
	FFBP neural network	Homoscedastic type PNN by PSO	Heteroscedastic type PNN by PSO	FFBP neural network	Homoscedastic type PNN by PSO	Heteroscedastic type PNN by PSO	FFBP neural network	Homoscedastic type PNN by PSO	Heteroscedastic type PNN by PSO
315 MVA	99.3	100	100	94.5	97.2	99.3	98.6	99.3	100
200 MVA	96.6	98.6	100	99.3	100	100	98.6	95.9	98.6
160 MVA	98.6	97.2	99.3	96.3	97.2	98.6	100	100	100

TABLE V
NUMBER OF POST DISTURBANCE SAMPLES REQUIRED FOR DECISION BY FFBP
NEURAL NETWORK AND PNN BASED RELAYS

Cases	Number of samples required (Actual)		Maximum Number of samples required (Logical)	
	FFBP	PNN	FFBP	PNN
Magnetizing inrush (0°)	11	05	12	12
Internal fault (Light phase-to-ground fault at 2%)	10	06	12	12

as the [1]: one cycle, [3]: between one and one-and-a-half cycle, [4]: with in one cycle, [5]: about 3/4 cycle, [6]: 10–13 ms (more than half cycle), [7]: between 12.4 ms to 19.6 ms (more than half cycle to about to one cycle). A few algorithms exists which have same fault detection time as the proposed method [23], [24]. However, these methods are based on wavelet which does not have satisfactory classification accuracy rate.

The proposed optimal PNN is capable to discriminate between magnetizing inrush and internal fault condition within half cycle. Table V shows that the PNN required less number of post disturbance samples to discriminate between the magnetizing inrush and internal fault conditions in comparison of FFBP neural network.

The proposed PNN is faster and have good generalization properties than the FFBP neural network. The training required for PNN is very different and much faster than that required for FFBP neural network. In PNN, the training process is one pass without any iteration for the weight adaptation, as against a large number of iteration (epochs) necessary in case of FFBP neural network. As example, 1000 iterations were required for convergence in case of 12-11-1 structure of FFBP neural network implemented by the authors, thereby giving a ratio of about 1000:1

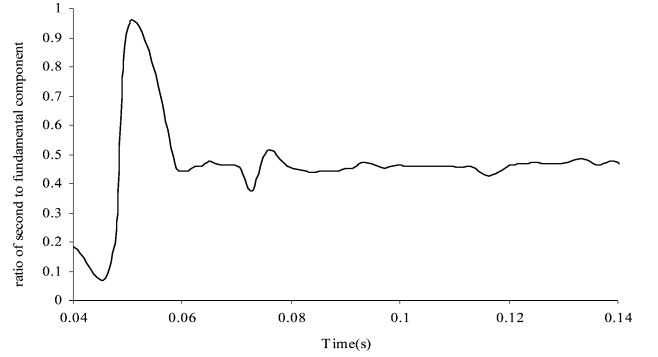


Fig. 12. Ratio of second harmonic to fundamental of the differential current under typical inrush condition (inrush occurs at 0.04 s).

in terms of the training time. In PNN there is fixed rule to decide the number of hidden layers and number of neurons in hidden layers as per application requirement whereas in case of RBFNN and FFBP neural network, architectures are decided empirically. This makes it easy to design and is simple in architecture than the classical ANN. PNN is influenced by smoothing factor only which can easily set by applying optimization technique like PSO. On other hand, to achieve good performance by classical ANN, two or more parameter have to be obtain empirically which is time consuming process. These characteristics of PNN make it ideal for real time application.

DFT-based harmonic restraint method is implemented, to compare performance of the proposed optimal PNN based algorithm in power transformer differential protection. Figs. 12–13 show the ratio of second harmonic to fundamental of the differential current under typical magnetizing inrush and internal fault conditions respectively. During one cycle under internal fault condition, the ratio of the second harmonic is quite high and in the same range as in case of magnetizing inrush condition. Therefore, it is difficult to discriminate between internal fault and inrush conditions merely setting a preset threshold. From Figs. 12–13, it is also clear that the ratio values are fluctuating, which create problem to decide a preset threshold. Moreover, due to the presence of second harmonic during internal fault condition digital relay will take longer time to make trip decision (one cycle or more than one cycle). In contrast, the optimal PNN based method is able to detect such a fault in 6 ms (half cycle or with in half cycle). However, the harmonic restraint method is capable to discriminate between these two conditions but does not seems to be intelligent to take decision in case of fluctuating ratio of second harmonic to fundamental of the differential current due to different loading conditions, severity of internal faults, switching-in angles etc. and hence mal-operation of relay will occur.

From the results, it is clear that the selection of an appropriate smoothing factor of PNN significantly enhances its performance and renders its classification accuracy. Tremendous capability of PNN for classification problems shows suitability for digital differential relaying protection scheme. It is free from the setting of threshold value. It is also immune from the different harmonics contained in operating signals which makes it simpler and robust than the conventional digital filtering algorithms.

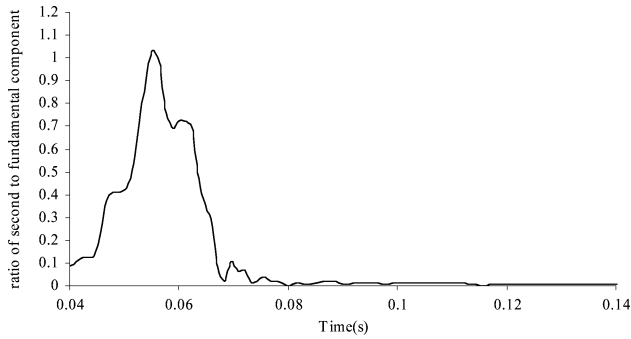


Fig. 13. Ratio of second harmonic to fundamental of the differential current under typical internal fault condition (internal fault occurs at 0.04 s).

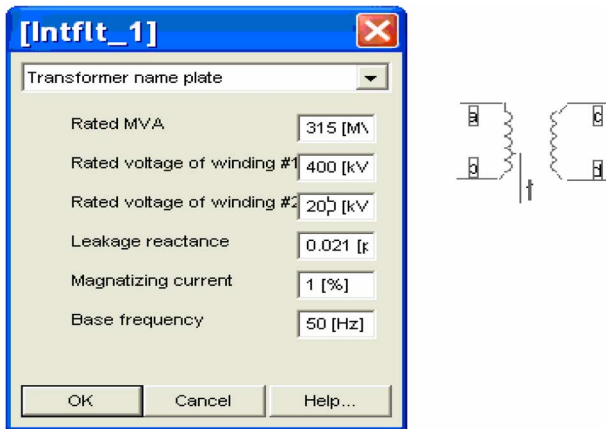


Fig. 14(a). Typical PSCAD/EMTDC transformer model to simulate internal fault.

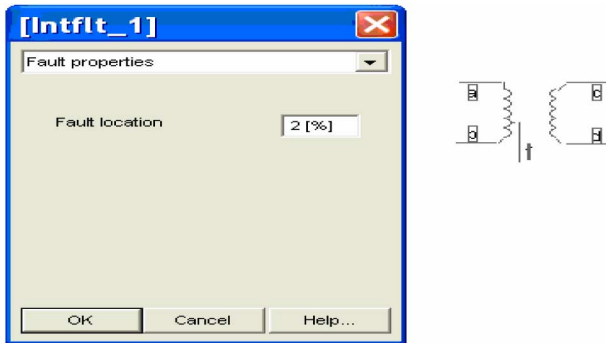


Fig. 14(b). Typical PSCAD/EMTDC transformer model to simulate internal faults at different locations.

VI. CONCLUSION

This paper presents a novel approach to discriminate between transformer internal fault and magnetizing inrush condition based on optimal PNN in digital differential protection scheme. In this article PSO technique is successfully applied to obtain the optimal smoothing factor of PNN model which is also easy to implement. From the results, it is clear that the selection of an optimal smoothing factor for PNN significantly enhances its performance.

PNN training process is one pass and without any iteration for weight adaptation, hence, yielding great processing speed as compare to FFBN neural network. The PNN generalizes to

the new incoming patterns without having to repeat the training process. These characteristics are ideal for real time applications.

The proposed optimal PNN algorithm is based on wave-shape identification technique and independent of the harmonics contained in differential current which is quite suitable in case of modern power transformers that use high-permeability low coercion core materials. The conventional harmonic restraint technique may fail because high second harmonic components are generated during internal faults and low second-harmonic components are generated during magnetizing inrush with such core materials. In the proposed method, stability of differential relay is ensured during the magnetizing inrush or sympathetic inrush, over-excitation, and external fault conditions. Hence, the differential protection reliability is enhanced.

APPENDIX I

Fig. 14 shows a typical PSCAD/EMTDC transformer model to simulate internal faults (turn-to-turn, phase-to-ground, and phase-to-phase) at different location of transformer winding from the neutral end of the windings. In this model MVA rating, voltage rating, base frequency, leakage reactance, magnetizing current, and fault location (in%), etc. can be defined.

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